SOFTWARE ENGINEERING CHECKLIST 1

Git Repository Verification Report

BUDDY – Intelligent Companion Robot Dog

Research Project ID	R26-SE-040
Specialization	Software Engineering
Institution	Sri Lanka Institute of Information Technology (SLIIT)
Supervisor	Pradeep K. W. Abeygunawardhana
Co-Supervisor	Samantha Rajapakse
Prepared for	Final Year Research Project – SE Checklist 1

Research Team

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01 Purpose and Repository Overview

This report provides evidence for the Software Engineering Checklist 1 requirement: Git repository. It demonstrates that the BUDDY research project is maintained in a structured GitHub repository with branch-based development, documented contribution rules, dependency files, deployment assets, configuration hygiene, and traceable commit history.

Repository: https://github.com/ashandth0309/R26-SE-040

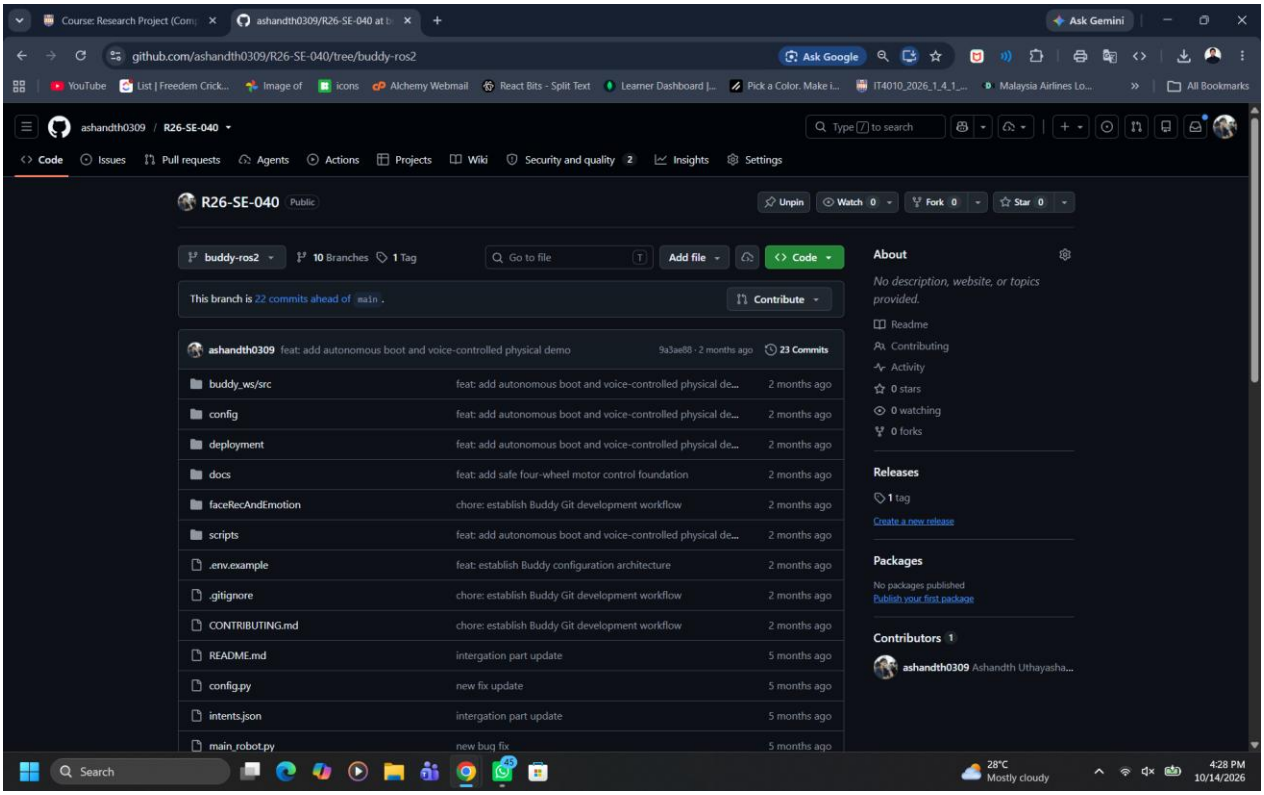


Figure 1 – BUDDY Git Repository Main Page

02 Repository Verification Summary

Requirement	Status
Public Git repository available	Verified
Branch-based development	Verified
Traceable commit history	Verified
Project README documentation	Verified / Updated
Repository structure and source code	Verified
Dependency file	Verified
Deployment configuration	Verified
Automated startup script/service	Verified
Contribution workflow	Verified
Secrets and generated-file exclusions	Verified
Environment variable example file	Verified

Assessment note: The repository uses separate branches for individual and integration work. Evidence below intentionally includes branch-specific histories rather than implying that all work occurred on a single branch.

03 Commit History and Parallel Development

The team used separate branches to preserve individual research contributions and integration work. The screenshots below show development histories for the three research streams.

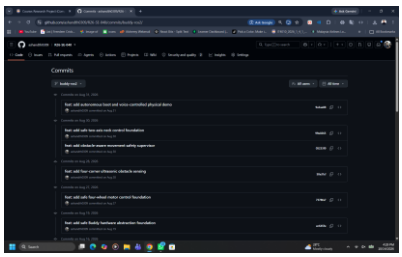


Figure 2A – Ashandth commit history (buddy-ros2)

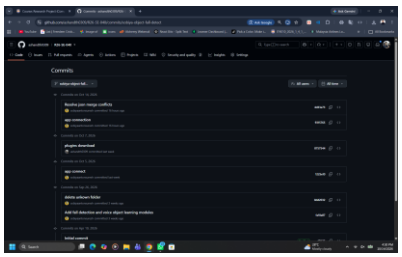


Figure 2B – Sobiya commit history (sobiya-object-fall-detect)

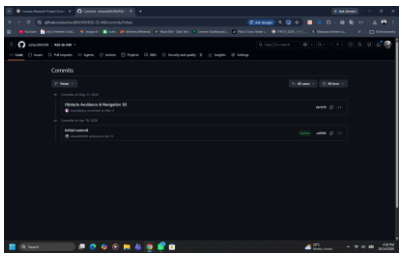


Figure 2C – Yohan commit history (Yohan)

- Ashandth: autonomous boot, voice-controlled physical demo, obstacle-aware safety supervision, ultrasonic sensing, motor control and hardware abstraction.
- Sobiya: app connection, merge resolution, visual/fall-detection related updates and supporting integration work.
- Yohan: obstacle avoidance and 3D navigation development on a dedicated branch.

04 Branch Structure and Project Documentation

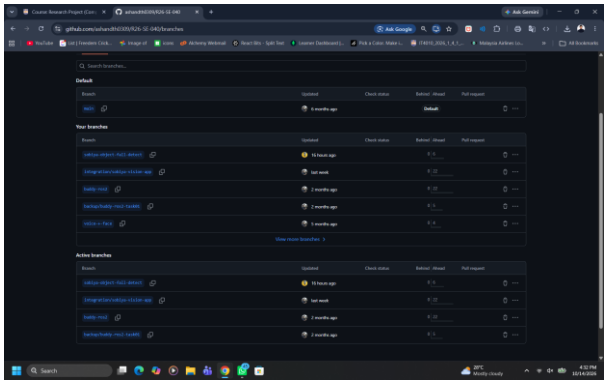


Figure 3 – Branch structure used for parallel development

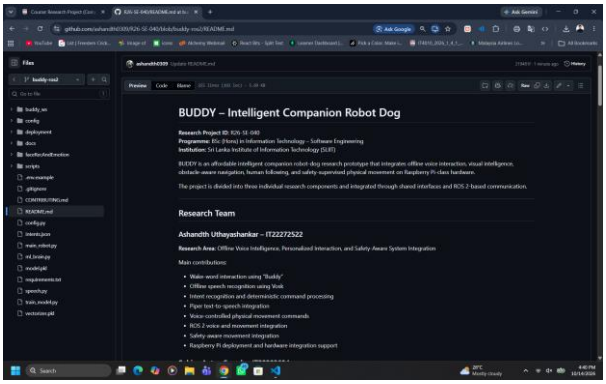


Figure 4A – Updated BUDDY README project overview

The repository contains dedicated branches for ROS 2 integration, Sobiya's vision/application work, integration snapshots, legacy work, and other feature development. The README was updated to represent the complete research project rather than only the earlier voice-assistant prototype.

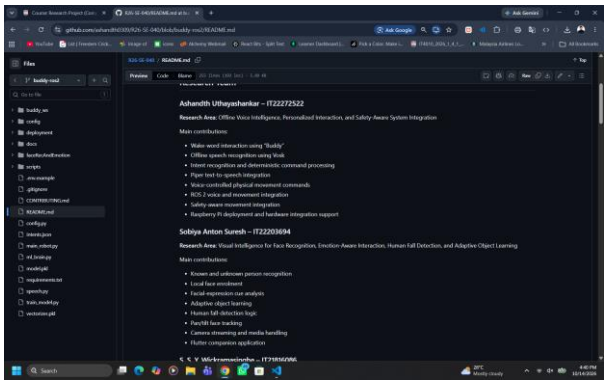


Figure 4B – Research team and individual contributions

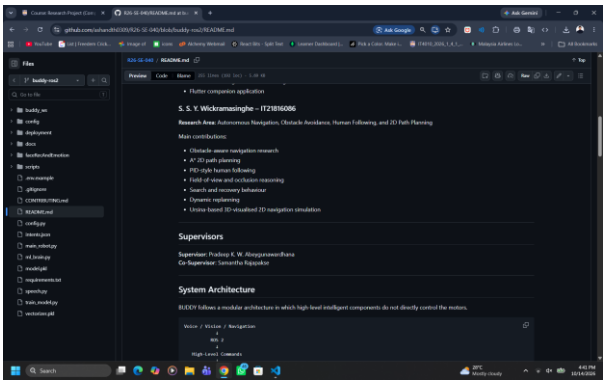


Figure 4C – Supervisors and system architecture

05 Repository Structure and Dependencies

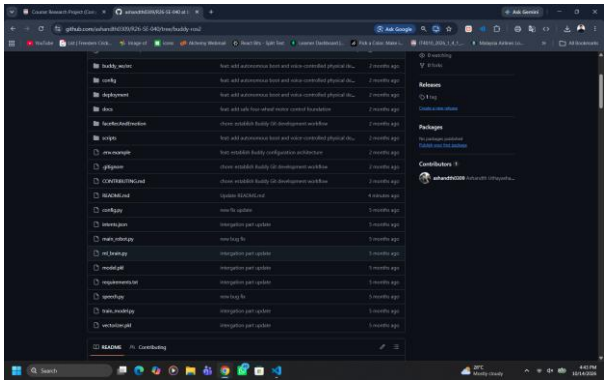


Figure 5 – BUDDY repository folder and file structure

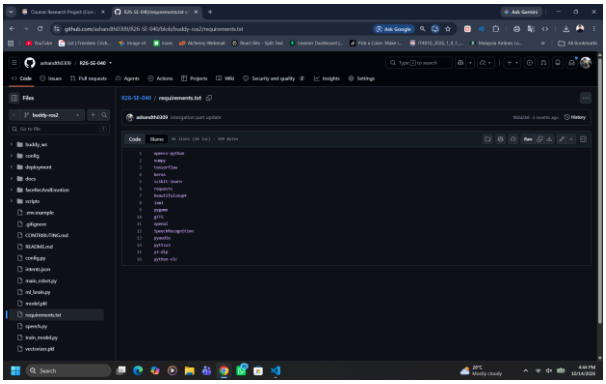


Figure 6 – Python dependency list (requirements.txt)

The repository structure includes ROS 2 source code, configuration, deployment assets, documentation, visual-intelligence code, startup scripts, environment examples, contribution guidelines and supporting Python modules.

Dependency note: requirements.txt contains a broader repository dependency snapshot, including packages used during earlier experimentation. It should not be interpreted as the exact final validated runtime dependency set for every subsystem.

06 Deployment Readiness and Automatic Startup

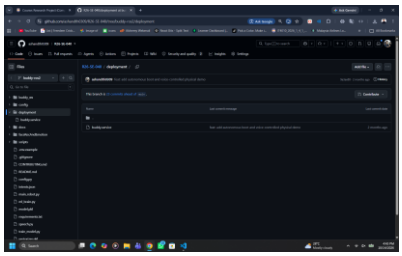


Figure 7 – Deployment folder containing buddy.service

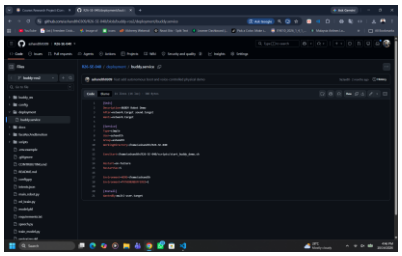


Figure 8 – systemd configuration for automatic BUDDY startup

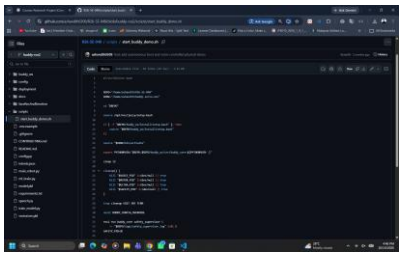


Figure 9 – start_buddy_demo.sh runtime startup script

- buddy.service defines the working directory, user/group, startup command, restart-on-failure policy and environment settings.
- start_buddy_demo.sh sources ROS 2 Jazzy, sources the built workspace when available, activates the Python environment, configures PYTHONPATH and starts coordinated robot processes.
- These files provide repository-level evidence that deployment/startup was considered as part of the engineering workflow.

07 Development Workflow and Repository Hygiene

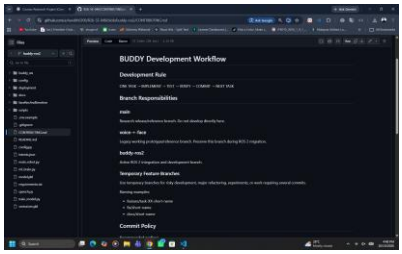


Figure 10 – Development workflow and branch responsibilities

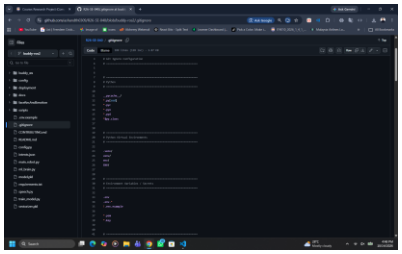


Figure 11 – .gitignore for generated files, environments and secrets

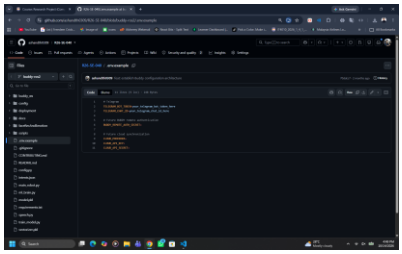


Figure 12 – .env.example safe configuration placeholders

- CONTRIBUTING.md documents a repeatable development cycle: implement, test, verify, commit and proceed to the next task.
- .gitignore excludes Python cache files, virtual environments, .env files and private-key file types.
- .env.example documents expected configuration keys using placeholders instead of real credentials.

08 Final Repository State

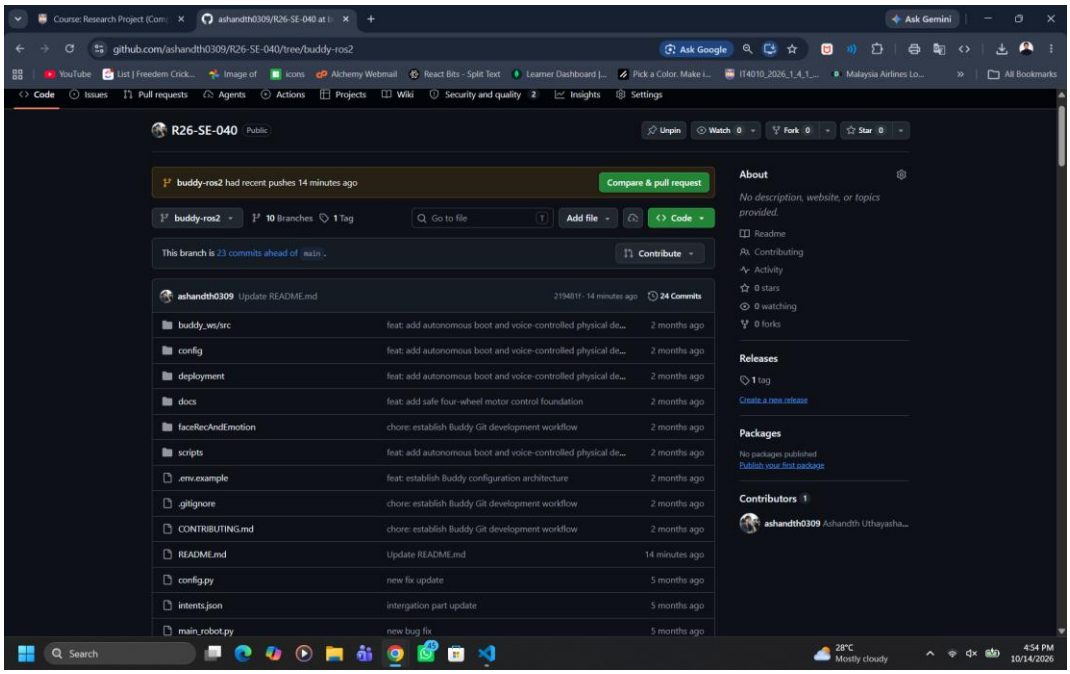


Figure 13 – Final BUDDY repository overview after documentation update

The final overview confirms that the active buddy-ros2 branch contains the updated README and major project assets, with a traceable commit history and dedicated development branches. The repository therefore satisfies the intended Software Engineering Checklist 1 evidence requirement for a managed Git repository.

09 Conclusion

The BUDDY repository demonstrates structured software-engineering practice through Git-based version control, branch separation, documented collaboration rules, reproducible configuration files, deployment scripts, and visible commit history. The evidence collected in this report supports the SE Checklist 1 requirement and provides a clear audit trail of the project's development and integration work.

Final status: **CHECKLIST 1 – GIT REPOSITORY: READY FOR SUBMISSION**